

Is There a Correlation Between a Country's Democracy Level and Visual Representation of Women in Diplomacy Related Media?

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MOTIVATION

When I deciding on my project, I wanted to work on a topic that interested me. When I saw The Diplomatic Text Dataset presented to us, I thought I could extract something about gender equality from the data.

I believe that, the media, in addition to being a factor that influences society, is a reflection of society. Moreover, I thought there might be a correlation between the representation of women in the media and the level of democracy in countries.



Data Source

For democracy index I used V-Dem democracy index,

"V-dem dataset (country-year: Full+others, v15)," Varieties of Democracy (V-Dem) Institute, 2025, accessed 2026-01-07. [Online]. Available: <https://www.v-dem.net/data/the-v-dem-dataset/country-year-v-dem-fullothers-v15/>

For the female ratios in the images I used the data set that is publicly provided by the VRL Lab, which is Diplomatic Text Dataset hosted on Harvard Dataverse,

"Diplomatic text dataset," Harvard Dataverse, 2024, globalDiplomacyNetproject dataset. Persistent ID: doi:10.7910/DVN/HYJDE0. [Online]. Available: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/HYJDE0>



Data Analysis

- Data construction
- Data filtering and aggregation
- Data integration
- Exploratory analysis
- Statistical analysis

- Data construction

Using Diplomatic text dataset, we calculated the total people and women seen in images and calculated the ratio. Moreover using the news IDs associated with the news images, these ratios adjusted to a yearly basis.

- Data Filtering and Aggregation

To prevent unidentifiable or misleading ratios, images where humans were not detected were excluded, and the analysis was limited to the period after 2000 due to the insufficient coverage of previous periods.

TABLE II PRE-2000 DATA COVERAGE IN THE CONSTRUCTED COUNTRY-YEAR PANEL.				
Year	<i>n_countries</i>	Rows	<i>people_total</i>	<i>n_images_people</i>
1950	1	1	10	1
1970	1	1	10	2
1998	1	1	1	1
1999	1	1	27	9

- Data Integration

The constructed panel was merged with V-Dem democracy indicators using country and year identifiers.

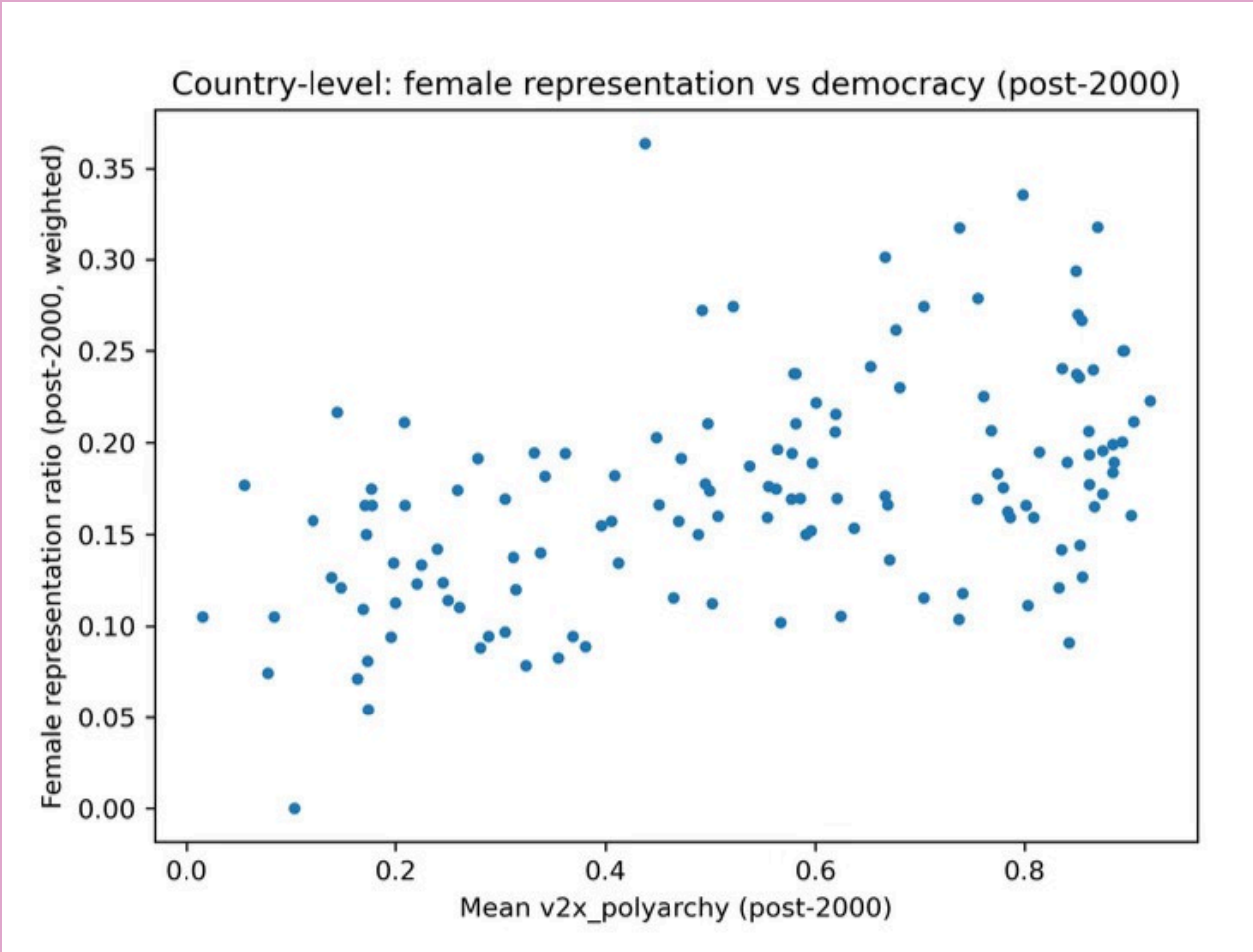
TABLE I SUMMARY OF CONSTRUCTED PANEL, MERGE COVERAGE, AND TEST SAMPLE (POST-2000).	
Statistic	Value
Country-year observations (all years)	1359
Countries observed (post-2000 panel)	143
Countries used in hypothesis test (post-2000)	137
Countries dropped (no V-Dem match post-2000)	6
Years covered (post-2000)	2000–2024
Share with <i>people_total</i> > 0	1.00
V-Dem match rate (<i>v2x_polyarchy</i> not missing)	95.36%

- Statistical analysis

A Spearman rank correlation was used to assess the monotonic relationship between democracy and female representation and statistical significance was evaluated using a permutation test to avoid parametric assumptions.

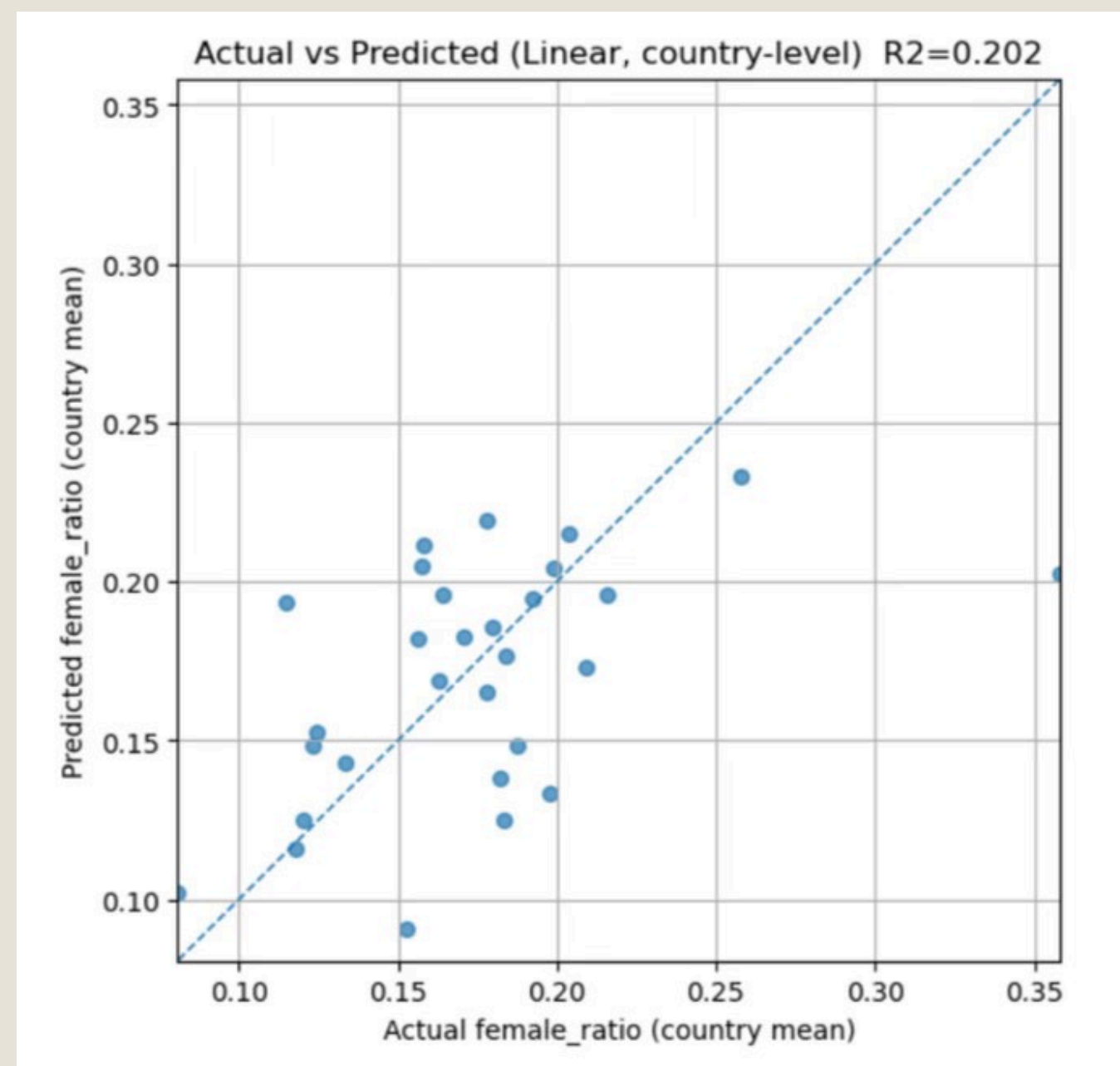
- Exploratory analysis

Time-series trends, country-level distributions, and scatter plots were used to explore patterns in female representation and democracy scores.

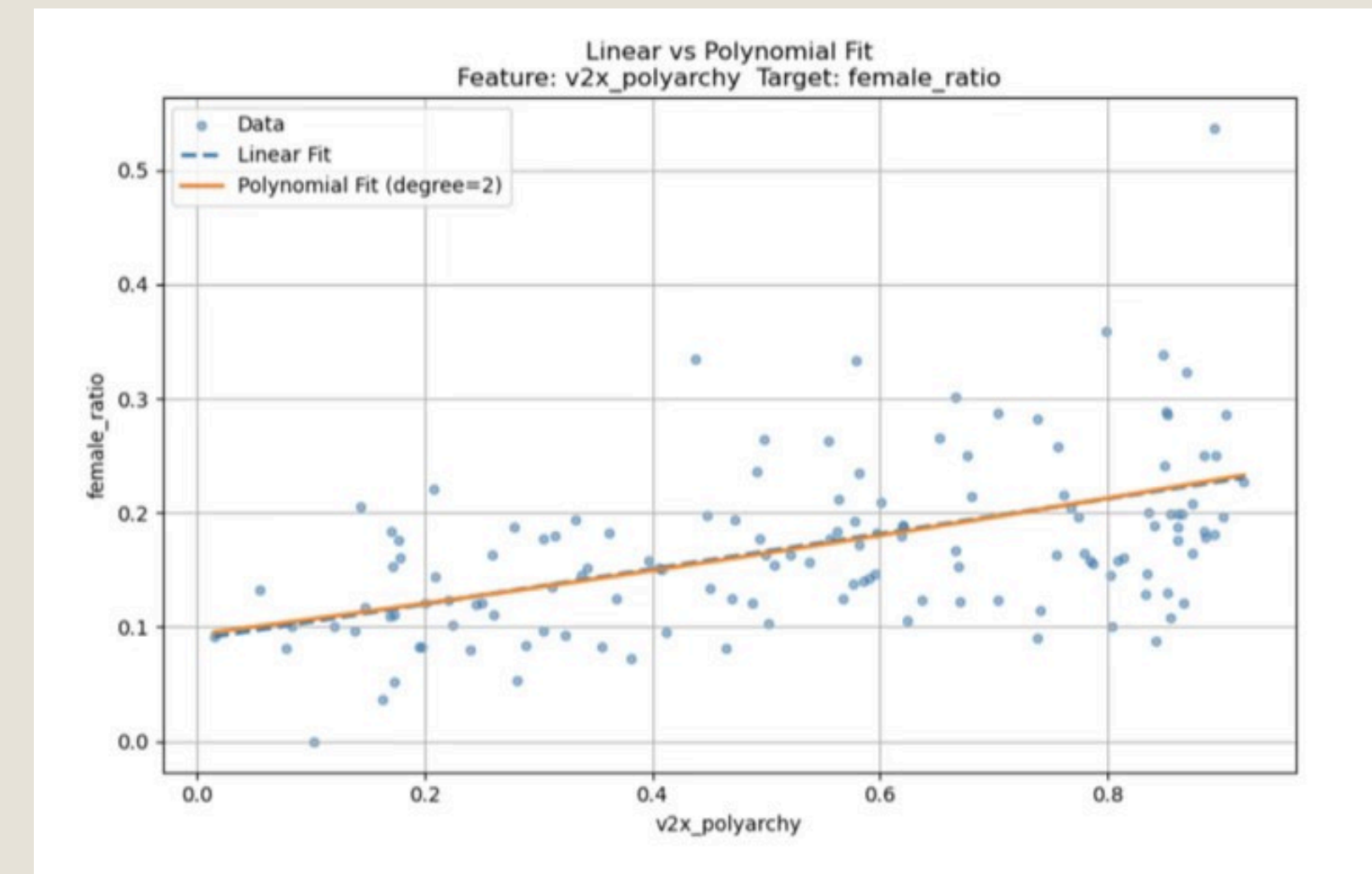


Findings

- As expected, a positive but moderate association is observed: countries with higher democracy scores tend to show higher female representation in diplomacy related imagery.



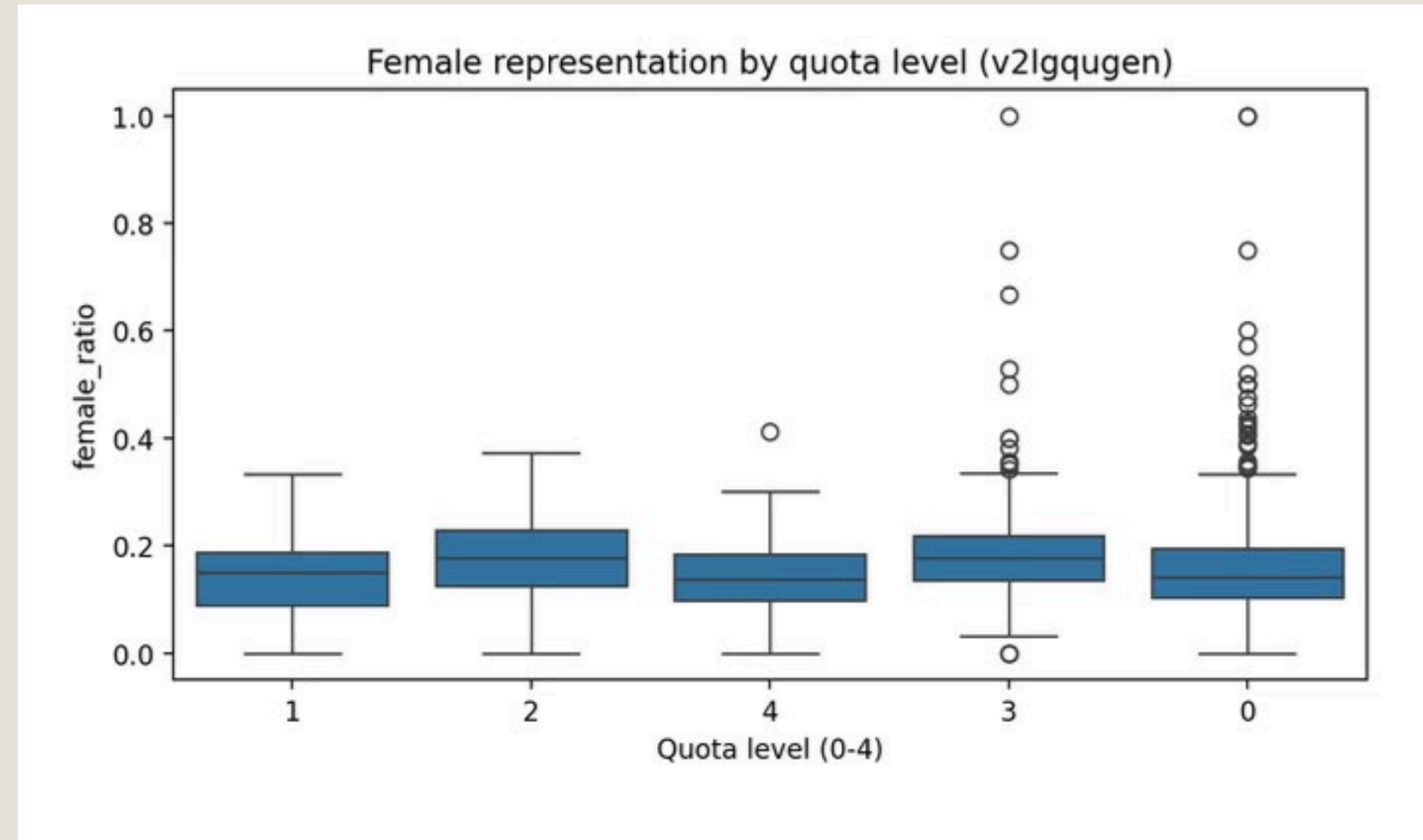
- Democracy alone explains limited variance ($R^2 \approx 0.20$).



- However, the relationship is moderate, so democracy is not a sole determinant of representation.
- There are high-democracy countries with relatively low female visibility, suggesting other factors also shape representation.

Findings

Additionally whether women's quota status is associated with female representation is also tested using a simple causal-inference setup.



- The regression-adjusted estimate was not statistically significant, so there is **no clear evidence of an effect** in this analysis.

Limitations

- **Limited sample size at the country level:**
The ML evaluation is conducted on country-level aggregates, which substantially reduces the number of observations.
- **Omitted cultural and societal factors:**
Institutional indicators may not capture relevant cultural determinants of gender visibility in media, such as religiosity, traditional gender norms, or broader societal attitudes toward women in public life.
- **Correlation, not causation:**
The analysis identifies association only. Causal interpretations cannot be made without stronger assumptions or causal designs.
- **Unobserved confounding (causal analysis):**
The causal interpretation relies on the assumption that the included covariates capture the main confounders jointly related to quota policy and female representation. Unmeasured factors (e.g., region, economic development, media ecosystem, diplomatic engagement intensity) may still bias estimated effects.

FUTURE WORK

Future studies could expand the feature set to include proxies for cultural and social norms. Specifically, creating a “traditional and modern” profile at the country level (e.g., through secularization proxies, gender norm indices, or survey-based measures) could help test whether such factors are related to women’s representation in diplomatic imagery and whether they interact with institutional democracy indicators.